

NovaMart Project Summary

NovaMart Data Warehouse — Project Summary

End-to-end cloud data warehouse for a multi-channel retailer, built by a 4-person team on AWS.

Problem

NovaMart, a mid-sized retailer selling through a website, a mobile app, and pop-up stores, had no single source of truth for sales performance. Finance, marketing, and operations each reported different numbers, and answering basic questions — which products are profitable, which channel is growing, how long fulfilment takes — meant days of manual spreadsheet work.

Approach

Designed and built a three-layer data warehouse on AWS (S3 → RDS PostgreSQL: staging → core → data mart), following a standard Kimball-style dimensional modelling process end to end:

- **Business requirements first** — translated a vague leadership complaint into 5 concrete, answerable business questions, classified as OLAP analytical queries against a chosen Sales business process.
- **Real-world data quality work** — profiled a real, messy public e-commerce dataset (re-themed as NovaMart data) and found, documented, and fixed 5 genuine data issues: empty-string dates, unresolved product categories, unstandardised text formatting, and a CSV-import tool bug that silently corrupted data.
- **Dimensional modelling** — designed a star schema at the atomic order-line grain, with correctly classified additive/non-additive facts, surrogate keys, a dummy-key strategy for unmatched foreign keys, and a role-playing Date dimension.
- **ETL & validation** — built and validated the staging-to-core load in SQL, proving referential integrity with foreign-key join checks (112,650 rows validated across every dimension).
- **Purpose-built data mart** — built a second, Finance-scoped mart with two fact table types: a periodic-aggregation revenue fact and an accumulating-snapshot order-fulfilment fact, justified in writing against the alternative.
- **BI dashboard** — connected AWS QuickSight directly to the mart and built a 5-visual dashboard, including a live non-additive profit-margin calculation and a visual on the accumulating snapshot.

Tools

AWS (S3, RDS PostgreSQL) · DBeaver · SQL · AWS QuickSight · GitHub

Outcome

A fully working, documented, portfolio-ready data warehouse: a live star schema with validated referential integrity, a Finance data mart with two fact table types, and a published BI dashboard answering the original business questions — plus a complete data dictionary, architecture diagram, ERD, and reproducible SQL scripts, ready to walk an interviewer through end to end.